

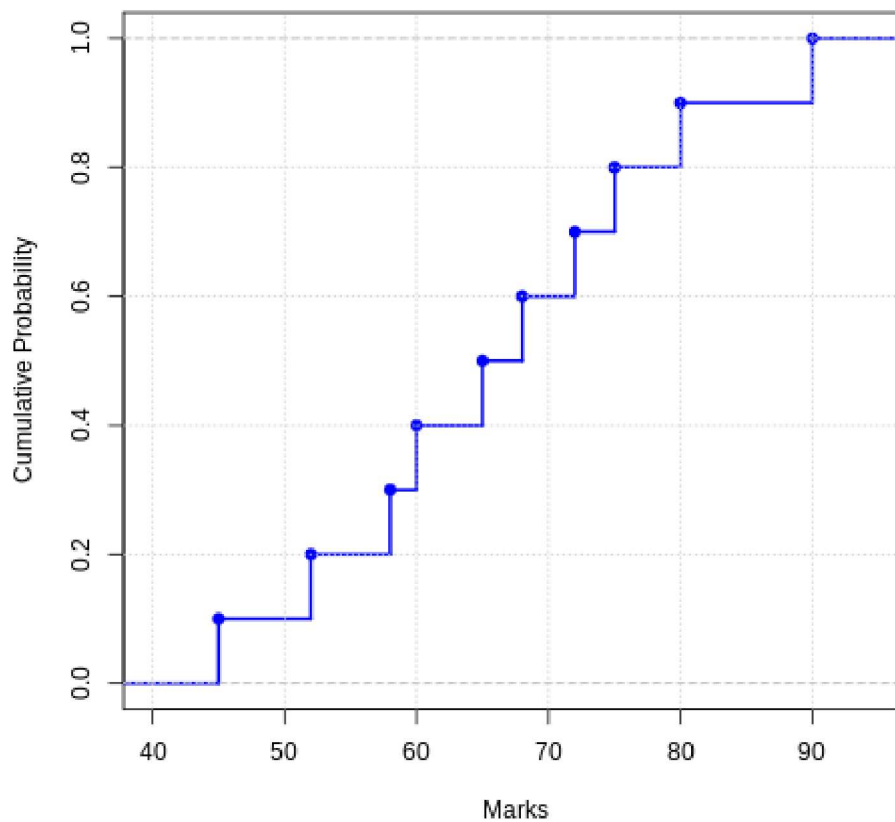
```
Processing triggers for man-db (2.10.2-1) ...  
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
```

```
!pip install rpy2  
%load_ext rpy2.ipython
```

```
Requirement already satisfied: rpy2 in /usr/local/lib/python3.12/dist-packages (  
Requirement already satisfied: cffi>=1.15.1 in /usr/local/lib/python3.12/dist-pa  
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages  
Requirement already satisfied: tzlocal in /usr/local/lib/python3.12/dist-package  
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packa  
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist  
The rpy2.ipython extension is already loaded. To reload it, use:  
%reload_ext rpy2.ipython
```

```
#1  
%%R  
  
marks <- c(45,52,58,60,65,68,72,75,80,90)  
  
ecdf_marks <- ecdf(marks)  
  
result <- data.frame(  
  Marks=sort(marks),  
  Cumulative_Probability=ecdf_marks(sort(marks))  
)  
  
print(result)  
  
plot(ecdf_marks,  
  main="ECDF of Student Marks",  
  xlab="Marks",  
  ylab="Cumulative Probability",  
  verticals=TRUE,  
  do.points=TRUE,  
  col="blue",  
  lwd=2)  
  
grid()
```

	Marks	Cumulative_Probability
1	45	0.1
2	52	0.2
3	58	0.3
4	60	0.4
5	65	0.5
6	68	0.6
7	72	0.7
8	75	0.8
9	80	0.9
10	90	1.0

ECDF of Student Marks

```
#2
%%R

steps <- c(4200,5000,5600,6100,6800,7200,8100,9200,10400,12000)

ecdf_steps <- ecdf(steps)

plot(ecdf_steps,
     main="ECDF of Daily Steps",
     xlab="Daily Steps",
     ylab="Cumulative Probability",
     verticals=TRUE,
     do.points=TRUE,
```

```
col="darkgreen",
lwd=2)

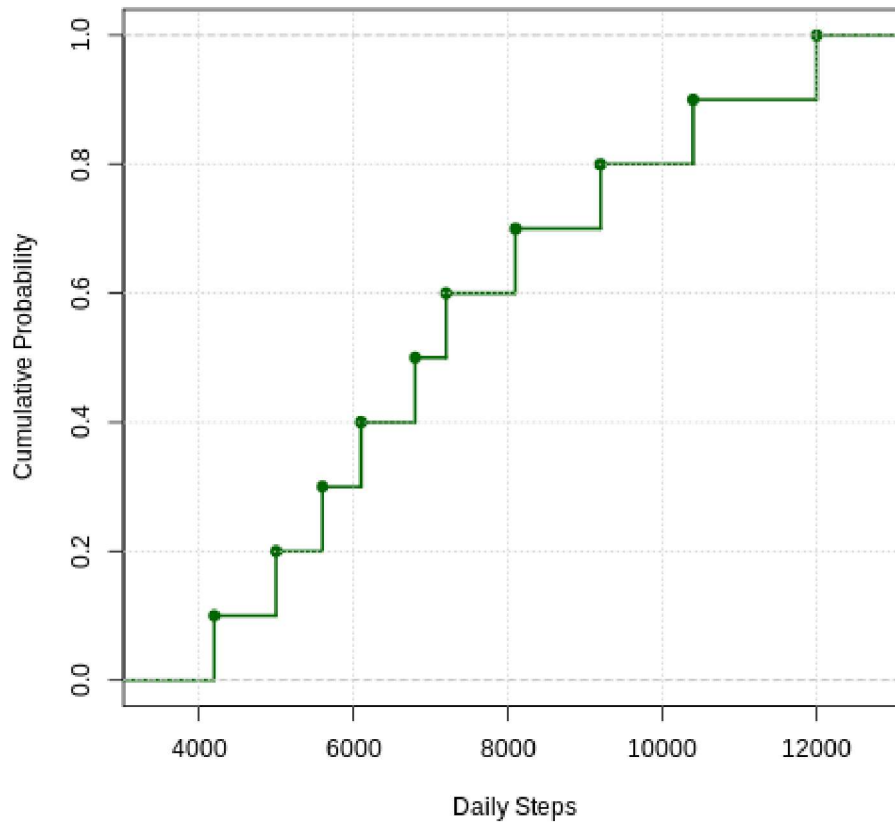
grid()

proportion <- ecdf_steps(8000)

cat("Proportion of participants with 8000 steps or fewer =", proportion,"\\n")
```

Proportion of participants with 8000 steps or fewer = 0.6

ECDF of Daily Steps



```
#3
%%R

purchase <- c(250,300,320,350,380,450,520,650,1800,5200)

hist(purchase,
probability=TRUE,
main="Histogram of Purchase Amount",
xlab="Purchase Amount",
col="lightblue",
border="black")

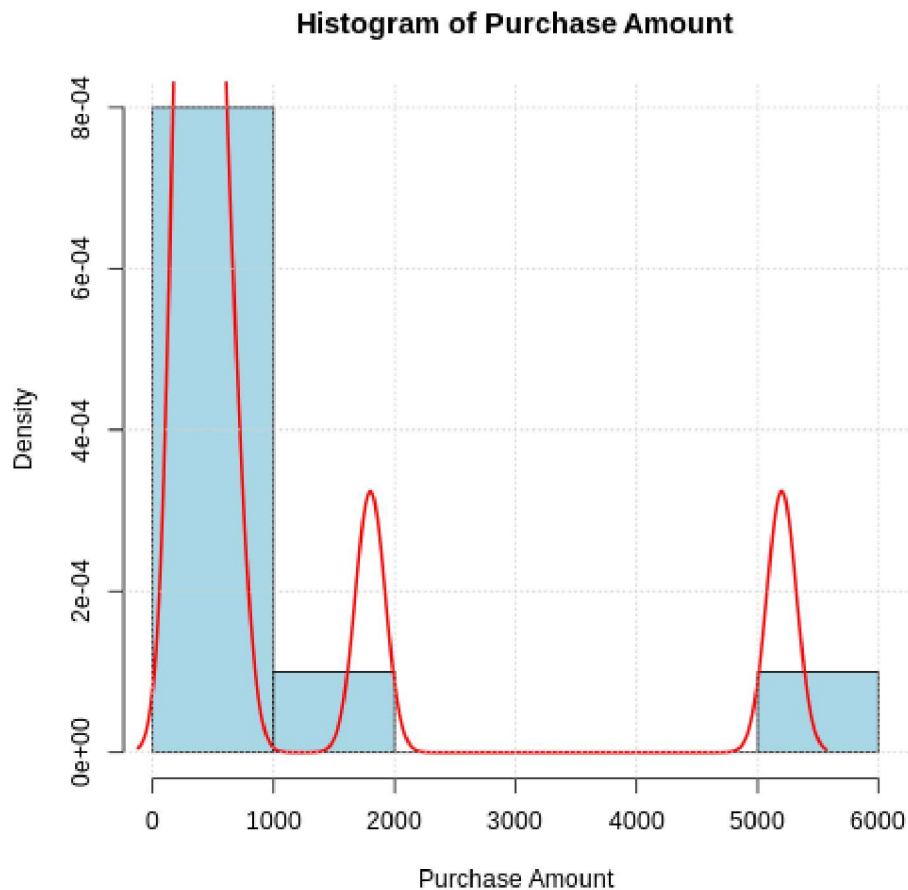
lines(density(purchase),
col="red",
```

```
lwd=2)

grid()

cat("Distribution is Positively Skewed\n")
```

Distribution is Positively Skewed



```
#4
%%R

waiting <- c(12,15,18,20,22,24,25,28,30,32)

qqnorm(waiting,
main="Normal Q-Q Plot of Waiting Time",
pch=19,
col="blue")

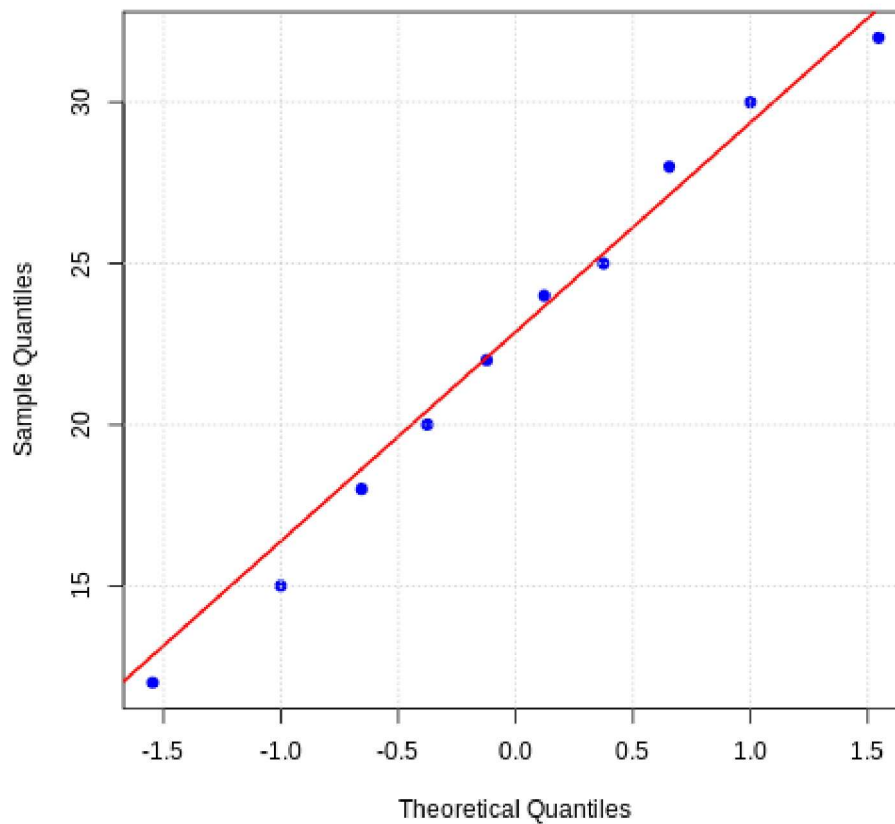
qqline(waiting,
col="red",
lwd=2)

grid()

cat("Waiting times approximately follow a normal distribution.\n")
```

Waiting times approximately follow a normal distribution.

Normal Q-Q Plot of Waiting Time



```
#5
%%R

income <- c(18000,22000,24000,26000,28000,
30000,34000,42000,85000,200000)

hist(income,
probability=TRUE,
main="Histogram of Monthly Income",
xlab="Monthly Income",
col="lightgreen",
border="black")

lines(density(income),
col="blue",
lwd=2)

grid()

qqnorm(income,
main="Normal Q-Q Plot of Monthly Income",
pch=19,
```

```
col="darkgreen")

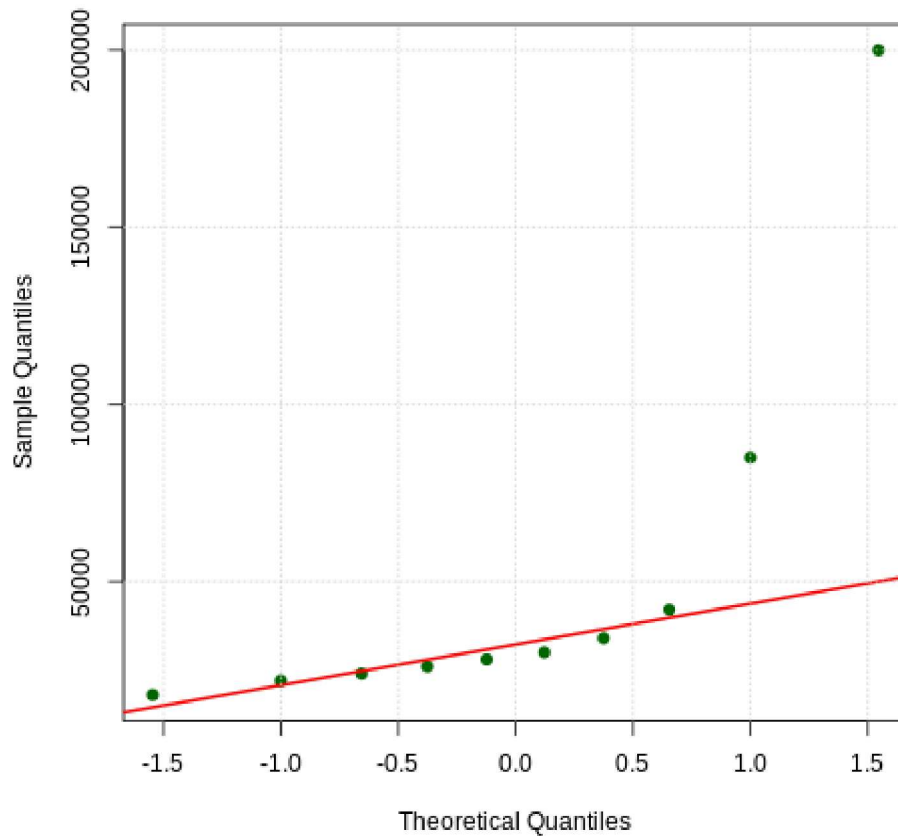
qqline(income,
col="red",
lwd=2)

grid()

cat("Income data is highly positively skewed and not normally distributed.")
```

Income data is highly positively skewed and not normally distributed.

Normal Q-Q Plot of Monthly Income



Start coding or [generate](#) with AI.